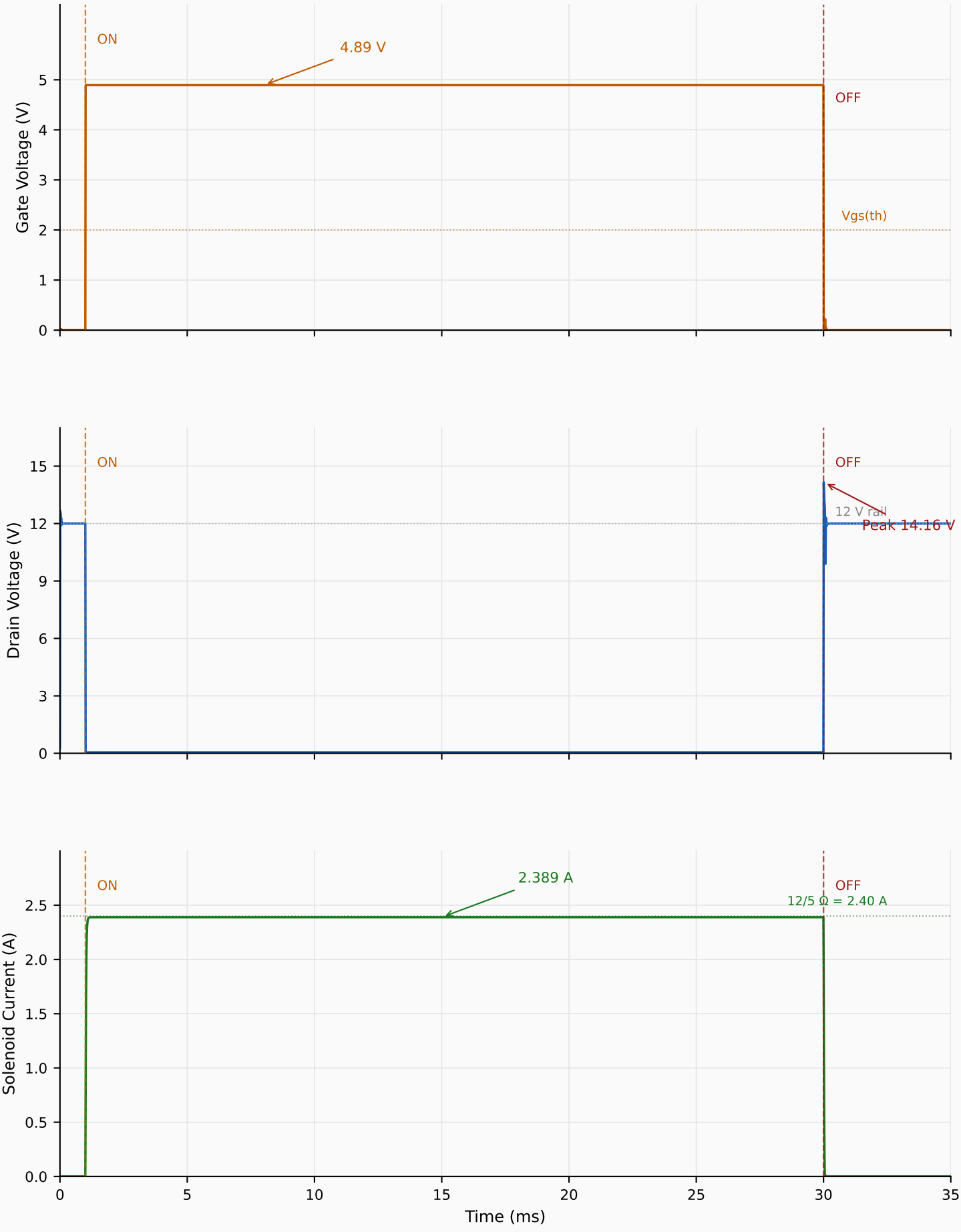
Q1 — Solenoid Drive Transient (IRLZ44N · 100 µH / 5 Ω · 12 V)



Q1 Solenoid Drive — Simulation Parameters

Parameter	Value	Notes
Flyback spike	**14.16 V**	+2.16 V above 12 V rail — within 55 V Vds(max)
Steady-state current	**2.389 A**	99.6 % of theoretical 12 V / 5 Ω = 2.400 A
Gate voltage	**4.89 V**	Vgs(th) ≈ 2 V — device fully enhanced at 5 V logic
L/R rise time (τ)	**20 µs**	100 µH / 5 Ω — 99 % settled in ~100 µs (5τ)
Rds(on) — IRLZ44N	**22 mΩ**	Datasheet value at Vgs = 5 V (corrected from gate current 22.7 m
P(Q1) = I²·Rds(on)	**125.6 mW**	2.389² × 0.022 Ω = 0.1256 W — negligible heating
MOSFET energy @ 45 s	**5.65 J**	0.1256 W × 45 s — well within package rating
MOSFET junction rise	**7.8 °C**	0.1256 W × 62 °C/W (TO-220, no heatsink) — Tj ≈ 33 °C vs 175
Solenoid coil power	**28.8 W**	I²R = 2.4² × 5 Ω — check solenoid duty-cycle rating
Solenoid energy @ 5 s	**144 J**	28.8 W × 5 s — verify winding thermal rating
Solenoid energy @ 45 s	**1296 J**	Worst-case — solenoid coil is the thermal limit, not the MOSFET

VERDICT: PASS. MOSFET thermal rise is <8 °C at any test duration. Verify the solenoid coil's energy/duty-cycle rating for the 5 s test run.